

Blocco 3

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In the following hf will stand for hyperfinite and hhf will stand for hyperhyperfinite.

Esercizio 1. Define an equivalence relation on the Baire space by setting xEy iff for all large n , $x_n = y_n$. Prove that E is hf.

Proof. Consider $f : \omega^\omega \rightarrow 2^\omega$ that maps $a \in \omega^\omega$ in the sequence obtained by the following steps:

1. Write the elements of a in binary.
2. Put each element from the previous step as a column of an $\omega \times \omega$ matrix
3. Scan the matrix diagonally (i.e. if M is the matrix, consider the sequence given by $\langle M(0,0), M(0,1), M(1,0), M(0,2), M(1,1), M(2,0), \dots \rangle$)

We claim f is a Borel reduction from E to E_0 . This will be enough because E_0 is hf.

- $xEy \Rightarrow f(x)E_0f(y)$: Let M, N the matrix associated to x, y respectively. Since xEy we have that $\exists k \in \omega \forall n \geq k \forall m (M(n, m) = N(n, m))$. Let $s \geq k$ such that

$$\forall t \geq s \forall r \leq k M(r, t) = N(r, t) = 0$$

Then $\forall n \geq \frac{2s \cdot (2s+1)}{2} (f(x)(n) = f(y)(n))$.

- $f(x)E_0f(y) \Rightarrow xEy$: Let $k \in \omega$ such that $\forall n \geq k (f(x)(n) = f(y)(n))$. Then

$$\forall n \geq k (x(n) = y(n))$$

□

Esercizio 2. Show that $E \leq_B E'$ and E' is hhf, then E is also hhf.

Proof. Since E' is hf, we can write $E' = \bigcup_{n < \omega} E'_n$ with $\langle E'_n : n < \omega \rangle$ increasing sequence of hf cbers. Let $f : X \rightarrow X$ be a Borel reduction from E to E' . For every $n < \omega$ define $x E_n y \iff f(x) E'_n f(y)$.

- $E \subseteq \bigcup_{n < \omega} E_n$:

$$xEy \Rightarrow f(x)E'f(y) \Rightarrow f(x)E'_n f(y) \text{ for some } n < \omega \Rightarrow x E_n y \Rightarrow x \bigcup_{n < \omega} E_n y \quad (1)$$

- $\bigcup_{n < \omega} E_n \subseteq E$:

$$x \bigcup_{n < \omega} E_n y \Rightarrow x E_n y \text{ for some } n < \omega \Rightarrow f(x)E'_n f(y) \text{ for some } n < \omega \Rightarrow f(x)E'f(y) \Rightarrow xEy \quad (2)$$

- for all $n < \omega$ E_n is hf cber :

Fix $n < \omega$. Then E_n is a cber because f, E'_n are Borel and $E_n \subseteq E$. E_n is hf because f is a Borel reduction from E_n to E'_n and E'_n is hf.

- $\langle E_n : n < \omega \rangle$ is increasing :

In fact $\langle E'_n : n < \omega \rangle$ is increasing, so we get

$$xE_ny \Rightarrow f(x)E'_nf(y) \Rightarrow f(x)E'_{n+1}f(y) \Rightarrow xE_{n+1}y \quad (3)$$

We conclude observing that $E = \bigcup_{n < \omega} E_n$ witnesses E hf.

□

Esercizio 3. Show that the Cohen forcing $Add(\omega, 1)$ is weakly homogeneous.

Proof. Let p, q in $Add(\omega, 1)$ and let $p' \leq p, q' \leq q$ such that $dom(p) = dom(q), p'(n) = 0$ if $n \notin dom(p), q'(n) = 0$ if $n \notin dom(q)$. Now define $\pi : Add(\omega, 1) \rightarrow Add(\omega, 1)$ by setting for all $r \in Add(\omega, 1)$ and for all $n \in dom(r)$

$$\pi(r)(n) = \begin{cases} p'(n) & \text{if } r(n) = q'(n) \\ q'(n) & \text{if } r(n) = p'(n) \\ r(n) & \text{else} \end{cases}$$

Observe that π is an automorphism that can be extended to $\bar{\pi} : \overline{Add(\omega, 1)} \rightarrow \overline{Add(\omega, 1)}$. We conclude since $\bar{\pi}(p') = q' \not\leq q'$.

□

For the next problem, let k be a measurable cardinal with a normal measure U and let P be the Prickry forcing at k .

Esercizio 4. Prove that P is weakly homogeneous.

Proof. WLOG we can assume we are considering the version of P where for all $(s, A) \in P$ we have $max(s) < min(A)$. Let $p, q \in P$. Let $p' \leq p, q' \leq q$ such that $|stem(p')| = |stem(q')| = l$ and $p' = (s, A), q' = (t, A)$. Let G generic such that $p' \in G$. We want to find H generic such that $q' \in H$ and $M[G] = M[H]$. If $\vec{\alpha}, \vec{\beta}$ are the generic sequences associated to G, H respectively, then $M[G] = M[H]$ is equivalent to $|\vec{\alpha} \Delta \vec{\beta}| < \omega$. So it is enough showing that there exists $\vec{\beta}$ generic sequence such that coincide with $\vec{\alpha}$ on a tail and such that $stem(q') \subseteq \vec{\beta}$.

Consider $\vec{\beta}$ defined as

$$\vec{\beta}(n) = \begin{cases} stem(q')(n) & \text{if } n \leq l \\ \vec{\alpha}(n) & \text{else} \end{cases}$$

- $\vec{\alpha}, \vec{\beta}$ coincide on a tail : This is directly from definition.
- $\vec{\beta}$ extends $stem(q')$: This is directly from definition.
- $\vec{\beta}$ is a generic sequence : Consider

$$H = \{(x, B) \in P : x \text{ is initial segment of } \vec{\beta}, \forall n \in \omega (\vec{\beta}(|x| + n) \in B)\}$$

Observe that G must be made in a similar fashion: $(y, C) \in G \Rightarrow y \subseteq \vec{\alpha}$. Furthermore $\forall n (\vec{\alpha}(|y| + n) \in C)$ because otherwise any element of G whose stem is $\vec{\alpha} \upharpoonright |y| + n$ (which must exist) would be incompatible with (y, C) .

It is clear that $\vec{\beta}$ is the generic sequence associated, but we need to verify H is generic. Let D dense and let $D' = \{(s \smallfrown z, C) : (t \smallfrown z, C) \in D\}$.

- $D' \neq \emptyset$ because D is dense.
- D' is dense under p' : In fact let $(s \smallfrown z, C) \leq p'$. By density of D there exists $(t \smallfrown z \smallfrown u, B) \in D$ with $B \subseteq C$. Then $(s \smallfrown z \smallfrown u, B) \in D'$.
- $G \cap D' \neq \emptyset$: In fact $p' \in G$, so one only needs to consider $\{r \in P : r \perp p'\} \cup D'$.

Let $(s \smallfrown z, C) \in G \cap D'$. Then:

- $(t \smallfrown z, C)$ is a valid condition because $\max(s) = \max(t)$
- $(t \smallfrown z, C) \in D$ because $(s \smallfrown z, C) \in D'$
- $(t \smallfrown z, C) \in H$ because $\vec{\beta}$ extends $t \smallfrown z$ and $\forall n (\vec{\beta}(|t \smallfrown z| + n) \in C)$ since $\vec{\beta}(|t \smallfrown z| + n) = \vec{\alpha}(|s \smallfrown z| + n) \in C$

So we have found $(t \smallfrown z, C) \in H \cap D$

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